MILITARY SPECIFICATION

STRAY VOLTAGE MONITOR ASSEMBLY

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains the Quality Assurance Provisions for inspecting and testing the Stray Voltage Monitor used in conjunction with the M-130 Aircraft General Purpose Dispenser.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-810 - Environmental Test Methods
MIL-STD-1460- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for

FSC: 4925

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.5)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9313794 - Stray Voltage Monitor Assembly
9322175 - Packing and Marking for Box, Exterior for Stray Voltage Monitor Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Officer, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. - ll?he components and assemblies shall comply wi'th all requirements specified on drawing 9313794 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Electrical tests

3.3.1 Circuit resistance. "- The electrical resistance of the circuits listed in Table I & II shall be 0.2 ohms maximum unless otherwise indicated when tested in accordance with 4.5.1.

3.3.2 Insulation resistance. - The insulation resistance of each isolated pin to all other pins (See Table III) electrically connected together shall be 20 megohms minimum when tested in accordance with 4.5.2.

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TABLE I

FIRE SELECT SEQUENCE SWITCH	FIRE SELECT RANGE SWITCH	FROM	TO
1	1-15	J1-A	P2 - Red lead
2	1-15	J1-B	P2 - Red lead
3	1-15	J1-C	P2 - Red lead
4	1-15	J1-D	P2 - Red lead
5	1-15	J1-E	P2 - Red lead
6	1-15	J1-F	P2 - Red lead
7	1-15	J1-G	P2 - Red lead
8	1-15	J1-H	P2 - Red lead
9	1-15	J1-J	P2 - Red lead
10	1-15	J1-K	P2 - Red lead
11	1-15	J1-L	P2 - Red lead
12	1-15	J1-M	P2 - Red lead
13	1-15	J1-N	P2 - Red lead
14	1-15	J1-P	P2 - Red lead
15	1-15	J1-R	P2 - Red lead
16	16-30	J1-S	P2 - Red lead
17	16-30	J1-T	P2 - Red lead
18	16-30	J1-U	P2 - Red lead
19	16-30	J1-V	P2 - Red lead
20	16-30	J1-W	P2 - Red lead
21	16-30	J1-X	P2 - Red lead
22	16-30	J1-Y	P2 - Red lead
23	16-30	J1-Z	P2 - Red lead
24	16-30	J1-a	P2 - Red lead
25	16-30	J1-b	P2 - Red lead
26	16-30	J1-c	P2 - Red lead
27	16-30	J1-d	P2 - Red lead
28	16-30	J1-e	P2 - Red lead
29	16-30	J1-f	P2 - Red lead
30	16-30	J1-g	P2 - Red lead

TABLE II

FROM	TO	WITH SELF TEST SWITCH	
J1-z	P2 (Red lead)	off	1.0 megohm min.
J1-z	P2 (Red lead)	Depressed	0.2 ohm max.
J1-BB	P1 (Black lead)	--	0.2 ohm max.

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INSULATION RESISTANCE

TABLE III

J1 - A, B, C, D, E, F, G, H, J, K, L, M, N, P, R, S, T, U, V, w, x, Y, Z, a, b, c, d, e, f, g, z, BB

3.4 Environmental requirements. - The Stray Voltage Monitc shall perform as specified in 3.3 after exposure to the environments specified herein.

3.4.2 Transportation vibration. - The Stray Voltage Monitc shall perform all functions satisfactorily after being subjected to the transportation vibration requirements of MIL-STD-810, Method 514.2, Procedure X. The Stray Voltage Monitor shall-be vibrated along three mutually perpendicular axes.

3.4.3 Temperature humidity. - The Stray Voltage Monitor shall perform all functions satisfactorily after being subjected to Test 105 of MIL-STD-331. Two fourteen day cycles shall be completed for a total test period of 28 days.

3.5 Wiring and soldering. - Soldering of electrical wires cables, connectors, components and assemblies shall be performed in a thorough, workmanlike manner in accordance-with MLL-STD-146

3.6 First article inspection. - This specification contains technical provisions for first article-inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract.

3.7 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required -markings shall be neat and sharply defined. Required packing shall be dry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the Contract. the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. - The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Stray Voltage Monitor	9313794	2

4.3.2 Inspections to be performed. - See MIL-A-48078 and paras 4.4.2.1 thru 4.4.2.6 testing per para. 4.4.3, and Table IV specified herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE IV - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63148(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Stray Voltage Monitor Assembly	1 OF 1	9313974
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Examination for defects	2	100%	3.2	4.4.2.4
	Circuit Resistance	2	100%	3.3.1	4.5.1
	Insulation Resistance	2	100%	3.3.2	4.5.2
	Transportation Vibration	2	100%	3.4.2	4.5.3
	Temperature & Humidity Cycling	2	100%	3.4.3	4.5.4
	Examination for defects	2	100%	3.2	4.4.2.4
	Circuit Resistance	2	100%	3.3.1	4.5.1
	Insulation Resistance	2	100%	3.3.2	4.5.2

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordante with MIL-STD-105, Inspection Level II Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63148(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Box	1 OF 1	9317943
			NEXT HIGHER ASSEMBLY 9313794

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any hole location incorrect		0.40%	3.2	Gage
102	Any hole size incorrect		0.40%	3.2	Gage
103	Lettering missing or incorrect		0.40%	3.2	Visual
Minor					
201	Protective finish missing, incorrect or damaged		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63148(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.2	Cover	1	1	9317942
				NEXT HIGHER ASSEMBLY: 9313794

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any hole location incorrect		0.40%	3.2	Gage
102	Any hole size incorrect		0.40%	3.2	Gage
103	Lettering missing or incorrect		0.40%	3.2	Visual
104	Screws missing or damaged		0.40%	3.2	Visual
Minor					
201	Protective finish missing, incorrect or damaged		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.7	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63148(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9317948
4.4.2.3	Meter Test Cable Assembly				NEXT HIGHER ASSEMBLY 9313794
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Banana Plug or wire color incorrect		0.40%	3.2	Visual
102	Overall length incorrect		0.40%	3.2	Gage
103	Insulation length incorrect		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.7	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-63148(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
4.4.2.4	Stray Voltage Monitor Assembly		1 OF 1		9313794
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Circuit resistance test		100%	3.3.1	4.5.1
102	Insulation resistance test		100%	3.3.2	4.5.2
103	Any component missing, damaged or incorrectly assembled		0.40%	3.2	Visual
104	Strain relief bushing missing or incorrectly assembled		0.40%	3.2	Visual
105	Soldering improper		0.40%	3.5	Visual
106	Insulation sleeving missing		0.40%	3.2	Visual
107	Lacing missing or insufficient		0.40%	3.2	Visual
108	Sequence switch pointer not aligned with markings		0.40%	3.2	Visual
109	Identification plate missing or incorrectly annotated		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.7	Visual

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4.4.3 Testing.- Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the “Paragraph Reference/ Inspection Method” column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). “The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and Section 6.3 herein.

4.5 Test methods and procedures -

4.5.1 Circuit resistance. - The requirements of 3.3.1 shall be verified by testing in accordance with MIL-STD-202, Method 303, using equipment in accordance with 4.4.4.

4.5.2 Insulation resistance. - The requirements of 3.3.2 shall be verified by testing in accordance with MIL-STD-202, Method 302, Test condition B, using equipment in accordance with 4.4.4.

4.5.3 Transportation vibration. - Using equipment in accordance with 4.4.4, subject each first article unit to the vibration requirements of MIL-STD-81O, Method 514.2, Procedure X to verify conformance to 3.4.2. Following this test, inspect each Stray Voltage Monitor for evidence of damage and submit each item to the procedures of 4.5.1 and 4.5.2. Any unit which fails to comply to the applicable requirements shall be rejected.

4.5.4 Temperature and humidity. - Using equipment in accordance with 4.4.4, subject each first article unit to test 105 of MIL-STD-331 to verify conformance to 3.4.3. Following this 28 day test, lower the chamber temperature to 130°F minimum and allow to stabilize for a minimum of 4 hours. Subject each unit to the electrical tests 4.5.1 and 4.5.2, at 130°F. The tests shall be completed within 30 minutes maximum after removal from the chamber. Following this test, place the units in a chamber at -60°F for a minimum of 4 hours and repeat the electrical tests. Any item which fails to meet applicable requirements shall cause the first article sample to be rejected.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be as specified on Dwg. 9322175.

5.2 Packing and marking. - Packing and marking shall be as specified on Dwg. 9322175.

6. NOTES

6.1 Intended use. - This Stray Voltage Monitor is used in testing the M-130 Aircraft General Purpose Dispenser.

6.2 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078). - Submit equipment designs as required, to Commander: U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards. - Distribution of data cards shall Include the following: Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAA-Q, Dover, New Jersey 07801.

6.5 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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